

A Dual Operator for Prime–Zero Coupling

and a Conditional Proof of Energy Asymmetry

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Abstract

We introduce the operator $\tilde{T} := \Phi \circ \Phi^*$, the dual of the loop operator $T = \Phi^* \circ \Phi$ constructed in Paper 3 of this series. Here $\Phi: H_{\text{str}} \rightarrow H_{\text{null}}$ maps a finite-dimensional prime space into a finite-dimensional zero space via Gaussian-damped prime resonance vectors.

What is proved. We establish (§ 2) the algebraic spectral identity $\sigma(T) \setminus \{0\} = \sigma(\tilde{T}) \setminus \{0\}$ by classical operator theory. We prove (§ 3.4) that $W_1 := C_T \cdot \tilde{T}^+$ is self-adjoint. W_1 does not assume RH or the HP postulate; however, its normalization constant C_T is calibrated against the zero ordinates via OLS and should not be presented as fully input-free. $\tilde{T}$ is explicit and deterministic; the γ_k are external numerical inputs, not conclusions of the model (§ 3.4). We prove (§ 4) the Abel Summation Principle (Lemma 4.3): for monotone decreasing weights f_i and a sequence g_i with bounded partial sums, $|\sum_i f_i g_i| \leq M f_1$. We prove (§ 4) the Prime Exponential Endpoint Estimate (Proposition 4.4): for each fixed zero ordinate γ_k ,

$$M_k(\kappa) := \max_{\substack{x \leq \kappa \\ p \leq x}} \left| \sum \sin(\gamma_k \log p) \right| \leq \left(\frac{1}{\sqrt{1 + \gamma_k^2}} + o_{\gamma_k}(1) \right) \pi(\kappa),$$

with no uniformity in k asserted, using the classical de la Vallée-Poussin form of the Prime Number Theorem, and no RH, GUE, or HP input.

What is numerical. The algebraic spectral identity is verified numerically: $\max_j |\lambda_j(T) - \mu_j(\tilde{T})| < 10^{-15}$ [Numerical, § 2.3]. At the benchmark grid $(\kappa, \varepsilon, N) = (53, 0.05, 100)$, the leading tested eigenvectors ($j = 1, \dots, 14$) of $\tilde{T}$ localize at zero ordinates with observed degrees 0.45–0.99 [Numerical, § 3.2]. The correlation $r_1 = \text{corr}(\mu_j, 1/\gamma_{k(j)}) = 0.950$ ($\kappa = 53$) [Numerical, § 3.1]. The energy asymmetry $\eta_{\text{ren}} > 0$ holds on the tested grid $\kappa \in \{23, 53, 101, 199, 503, 1009\}$ [Numerical, § 5.1]. The operator $\tilde{T}$ is not supported as a Hilbert–Pólya operator by the tested spectral diagnostics: $r_2 = \text{corr}(\omega_j, \gamma_{k(j)})$ falls from 0.50 to 0.16 on the tested grid $\kappa \in \{23, \dots, 503\}$ [Numerical, § 2.4].

What is conditional. Under Assumption (E_{rem}) (remainder control: $|\Delta_{\text{Cross}} + \Delta_{\text{Stream}}| \leq \rho \cdot \Delta_{\text{Burst}}$ for some $\rho < 1$, numerically $\rho_{\text{max}} = 0.583 < 1$ on the full tested grid $\kappa \in \{23, 53, 101, 199, 503, 1009\}$) we prove (§ 5.3) that $\eta_{\text{ren}}(\kappa) \geq (1 - \rho)\Delta_{\text{Burst}}/E_{\text{str}}(\kappa) > 0$ [conditional, by triangle inequality]. We establish numerically that $\Delta_{\text{Burst}} = 3.105\dots > 0$ is κ -invariant by definition for $\kappa \geq 7$ at fixed $(\varepsilon, N) = (0.05, 100)$. On the tested grid $\kappa \in \{23, 53, 101, 199, 503, 1009\}$, one observes $\Delta(\kappa) \geq \Delta_{\text{Burst}}$; analytically, this stronger lower-bound property is OPEN. Under Assumption (E_{rem}), the conditional theorem gives $\Delta(\kappa) \geq (1 - \rho)\Delta_{\text{Burst}} > 0$, and on the tested grid $\kappa \in \{23, 53, 101, 199, 503, 1009\}$, the values of $\eta_{\text{ren}}(\kappa)$ are positive [numerical extrapolation; no convergence theorem].

What remains open. The analytic proof of $\eta_{\text{ren}} > 0$ without the above assumption (§ 6); whether a circularity-free spectral function $f(\tilde{T})$ exists with spectrum converging to $\{\gamma_k\}$ (§ 6); and the arithmetic origin of $C_\eta \approx 0.39$ (§ 6).

Structure. The algebraic statements in §§ 2–3 are non-circular and proof-complete; the spectral diagnostics in § 3 are numerical. § 4 contains proved estimates; § 5 contains the numerical and conditional energy-asymmetry analysis. Section 6 states the open problems. No use of the Riemann Hypothesis, the GUE conjecture, the Montgomery pair correlation conjecture, or the Hilbert–Pólya postulate is made anywhere in §§ 2–5.

Notation and Conventions

Critical line. Throughout this series, the *critical line* means $\Re(s) = \frac{1}{2}$, parameterised by $s = \frac{1}{2} + i\gamma$ with $\gamma \in \mathbb{R}$.

Global parameters. The benchmark parameter set is $\kappa = 53$ ($\pi(\kappa) = 16$ active primes), $\varepsilon = 0.05$, $N = 100$ Riemann zero ordinates, $\sigma_0 = \frac{1}{2}$. Numerical sections explicitly state when κ is varied; $\varepsilon = 0.05$ and $N = 100$ are fixed unless stated otherwise.

Main objects. The encoding map $\Phi: H_{\text{str}} \rightarrow H_{\text{null}}$ assigns to each prime $p \leq \kappa$ a resonance vector $a_p = (e^{-\varepsilon^2 \gamma_k^2 / 2} \sin(\gamma_k \log p))_{k=1}^N$. The loop operator $T = \Phi^* \circ \Phi$ acts on the prime space H_{str} ; its dual, the Tehrani operator $\tilde{T} = \Phi \circ \Phi^*$, acts on the zero space H_{null} and is the central object of this paper. The canonical weight $c_p^{\text{ren}} := \sqrt{f_p}$ is the Weil-renormalised weight from Paper 2, § 4. The energy asymmetry is $\eta_{\text{ren}} := \eta[c^{\text{ren}}] = 1 - c^T G^{\text{un}} c / E_{\text{str}}$, where $\eta[c]$ denotes the asymmetry functional evaluated at weight c . Since all Paper-4 energy computations are at the fixed reference line $\sigma_0 = \frac{1}{2}$, we write $\eta_{\text{ren}}(\kappa)$ for $\eta[c^{\text{ren}}]$ at fixed $(\sigma_0, \varepsilon, N) = (\frac{1}{2}, 0.05, 100)$. This differs from Paper 3’s $\eta_{\text{orig}} := \eta[c^\eta]$, which uses $c_p^\eta = \sqrt{V_p(\frac{1}{2})}$. At $\kappa = 53$ and fixed $(\sigma_0, \varepsilon, N) = (\frac{1}{2}, 0.05, 100)$, the two reference values are

$$\eta_{\text{ren}} = 0.6693, \quad \eta_{\text{orig}} = 0.6908.$$

Two arithmetic constants govern spectral scaling: $C_\eta \approx 0.39$ (renormalised eigenvalue ratio, Definition 3.2) and C_T (OLS slope of μ_j vs. $1/\gamma_{k(j)}$).

Analytical foundations. The proofs use the Prime Number Theorem (Hadamard 1896, de la Vallée-Poussin 1896), the Abel partial summation formula, standard properties of Moore–Penrose pseudoinverses, and standard finite-dimensional spectral theory, including the singular-value identity for $\Phi^* \Phi$ and $\Phi \Phi^*$.

Numerical computations. All numerical results were computed with `mpmath` at 50 decimal digits of precision, using `mpmath.zetazero` for Riemann zero ordinates. Matrix eigendecompositions and pseudoinverses were computed using NumPy (`numpy.linalg`) with standard double-precision floating point; the numerical rank of $\tilde{T}$ is determined by a threshold of 10^{-10} relative to the largest singular value. Verification scripts are available at <https://github.com/utehrani/analysislab-nt>.

Not to be confused with. $C_\eta \neq C_T$: C_η denotes the observed finite-grid ratio $R_\eta(\kappa) =$

$\lambda_{\text{max,ren}}/\pi(\kappa) \approx 0.39$ (no limit theorem is asserted), while C_T governs the finite-grid OLS fit $\mu_j \approx C_T(\kappa)/\gamma_{k(j)}$. The inverse operator $W_1 = C_T \cdot \tilde{T}^+$ is *not* a Hilbert–Pólya operator; its HP-correlation r_2 decays to 0.16 with growing κ . The renormalised asymmetry η_{ren} uses weight $c_p^{\text{ren}} = \sqrt{f_p}$; Paper 3’s η_{orig} uses $c_p^\eta = \sqrt{V_p(\frac{1}{2})}$: these are numerically close but conceptually distinct.

1 Introduction

1.1 Background and Motivation

This paper is the fourth in a series developing an operator-theoretic framework for studying the distribution of the non-trivial zeros of the Riemann zeta function. The operator-theoretic algebra in §§ 2–3 is non-circular and proof-complete; the spectral diagnostics in § 3 are numerical. § 4 contains proved estimates; § 5 contains the numerical and conditional energy-asymmetry analysis.

Paper 1 [17] introduced the local curvature function and established the divergence $H_{\text{local}}(\frac{1}{2}, \kappa) \sim 2(\log \kappa)^2 \rightarrow \infty$ at $\sigma = \frac{1}{2}$.

Paper 2 [18] connected this local curvature to the Weil explicit functional, constructing admissible test functions for which the curvature divergence becomes a structural feature of the Weil energy.

Paper 3 [19] made the geometric structure explicit: two finite-dimensional real Hilbert spaces H_{str} and H_{null} connected by a linear map Φ , with loop operator $T = \Phi^* \circ \Phi$ and numerical energy asymmetry for the Paper 3 weight $c_p^\eta = \sqrt{V_p(\frac{1}{2})}$ (denoted η_{orig} in Paper 3). These results appear in § 1 for context; the formal core §§ 2–5 depends only on the definitions in § 2.1.

The present paper introduces and analyzes $\tilde{T} = \Phi \circ \Phi^*$, the dual of T , acting on the zero space H_{null} . The operator emerges canonically from the same Φ and requires no new postulate. Several established programs seek a spectral interpretation of the Riemann zeros; we position the present work relative to the main ones.

A Hilbert–Pólya operator $\mathcal{H}$ would satisfy $\mathcal{H}\psi_k = \gamma_k\psi_k$; the zero ordinates appear directly as eigenvalues. The Tehrani operator $\tilde{T}$ has eigenvalues $\mu_j \approx C_T(\kappa)/\gamma_{k(j)}$ (finite-grid OLS fit, numerically); the zero ordinates appear as localization centers of eigenvectors, not as eigenvalues. We use the standard formulation of the Hilbert–Pólya setting (see e.g. the survey by Bombieri [2] and the physics review [13]) without adopting it as a postulate. We call $\tilde{T}$ a “resonance operator” to express eigenvector concentration around arithmetic centers γ_k ; this is paper-specific terminology, not used in the scattering-theoretic sense.¹

Berry and Keating [1] seek a self-adjoint operator whose eigenvalues are γ_k via xp -quantisation; the present paper operates in the opposite direction (γ_k as eigenvector centers, not eigenvalues). Burnol [4] constructs complete and minimal systems in Sonin spaces indexed by primes and zeros via a conductor operator, the closest published antecedent to the $H_{\text{str}}-H_{\text{null}}$ duality; the present paper differs in using a finite-rank linear map Φ with explicit Gaussian phasors rather than an infinite-dimensional Fourier-analytic conductor operator. Connes [6] gives an adelic trace-formula interpretation; no formal connection to $\tilde{T}$ is established here; for Weil positivity at the archimedean place, see Connes–Consani [7]. Sierra and Townsend [15] provide a truncated

¹We call $\tilde{T} = \Phi\Phi^*$ the *resonance operator* (Tehrani operator), following the notation of this series. This should not be confused with resonance operators in the Lax–Phillips scattering theory or with “resonances” in the Sierra–Rodríguez-Laguna model [14], where the term refers to poles of the resolvent. Here $\tilde{T}$ is a positive semidefinite Gram-type operator on the finite-dimensional space H_{null} . For analogous HP falsification results, see [8].

Hamiltonian realization connected to Connes’ absorption spectrum; $\tilde{T}$ is not a Hamiltonian but a dual encoding operator. The Bost–Connes system [3] operates in a quantum-statistical framework with KMS states; the present construction is purely finite-dimensional and linear-algebraic. Most recently, Connes–Consani–Moscovici [5] construct a self-adjoint operator as a rank-one perturbation of the scaling spectral triple, achieving numerical alignment with the first fifty zero ordinates; our construction differs structurally: $\tilde{T} = \Phi\Phi^*$ is a Gram-type operator on H_{null} , indexed by sin-phase prime phasors, without spectral postulates and without rank-one perturbation structure. The estimate, for each fixed γ_k , $M_k(\kappa) \leq (1/\sqrt{1 + \gamma_k^2} + o_{\gamma_k}(1))\pi(\kappa)$ (no uniformity in k) is a standard prime exponential sum bound proved by PNT/Abel-summation methods; see also Turán’s program [16]. The closest historical precedent for the oscillatory-sum estimate is Turán’s [16] program on approximative Dirichlet polynomials; the novelty of our Proposition 4.4 lies in how it may enter a future analytic route to the energy-asymmetry problem.

The main conditional result is a proof that $\eta_{\text{ren}} > 0$ under Assumption (E_{rem}) — that the cross/stream energy remainder $|\Delta_{\text{Cross}} + \Delta_{\text{Stream}}|$ is bounded by $\rho \cdot \Delta_{\text{Burst}}$ for some $\rho < 1$. Numerically $\rho_{\text{max}} = 0.583 < 1$ on the full tested grid $\kappa \in \{23, 53, 101, 199, 503, 1009\}$.

1.2 Main Results

We prove the algebraic spectral identity $\sigma(T) \setminus \{0\} = \sigma(\tilde{T}) \setminus \{0\}$ (§2), self-adjointness of $W_1 = C_T \cdot \tilde{T}^+$ (§3.4), the Abel Summation Principle (Lemma 4.3, §4.3), and the Prime Exponential Endpoint Estimate (§4.4, Proposition 4.4) using only the Prime Number Theorem. We establish numerically that $\tilde{T}$ is not a Hilbert–Pólya operator (§2.4), that $\eta_{\text{ren}} > 0$ on the tested grid $\kappa \in \{23, 53, 101, 199, 503, 1009\}$ (§5.1), and that $\Delta_{\text{Burst}} = 3.105 > 0$ is κ -invariant by definition for $\kappa \geq 7$ (§5.2); $\Delta(\kappa) \geq \Delta_{\text{Burst}}$ is observed on the tested grid but not proved without additional assumptions. Under Assumption (E_{rem}), $\eta_{\text{ren}}(\kappa) \geq (1 - \rho) \Delta_{\text{Burst}}/E_{\text{str}}(\kappa) > 0$ (§5.3).

1.3 Reader Contract

This paper *proves*, by classical operator theory and the Prime Number Theorem, the spectral identity $\sigma(T) \setminus \{0\} = \sigma(\tilde{T}) \setminus \{0\}$, self-adjointness of W_1 , the Abel Summation Principle, and, for each fixed γ_k , $M_k(\kappa) \leq (1/\sqrt{1 + \gamma_k^2} + o_{\gamma_k}(1))\pi(\kappa)$ with no uniformity in k (Proposition 4.4). These are unconditional.

This paper *establishes numerically* that $\tilde{T}$ is not a Hilbert–Pólya operator (r_2 falls to 0.16 on the tested grid), that $\eta_{\text{ren}} > 0$ on the tested grid $\kappa \in \{23, 53, 101, 199, 503, 1009\}$, and that $\Delta_{\text{Burst}} = 3.105 > 0$ is κ -invariant by definition for $\kappa \geq 7$. These are verified on a finite grid and labelled [numerical].

This paper *proves conditionally/numerically*, under Assumption (E_{rem}) (§5.3) and the finite numerical verification $\Delta_{\text{Burst}} > 0$ at $(\varepsilon, N) = (0.05, 100)$, that $\eta_{\text{ren}}(\kappa) \geq (1 - \rho) \Delta_{\text{Burst}}/E_{\text{str}}(\kappa) > 0$.

This paper *leaves open* seven problems: rank equality for $\tilde{T}$ (Problem 6.1), the analytic proof

of Assumption (E_{rem}) (Problem 6.2), the analytic proof of $\eta_{\text{ren}} > 0$ without assumptions (Problem 6.3), a circularity-free HP spectral function $f(\tilde{T})$ (Problem 6.4), the arithmetic origin of $C_\eta \approx 0.39$ (Problem 6.5), the asymptotics of $C_T(\kappa)/\pi(\kappa)$ (Problem 6.6), and whether $\eta_{\text{ren}} > 0$ follows from PNT alone (Problem 6.7).

2 The Tehrani Operator

2.1 Recollection: H_{str} , H_{null} , Φ , T

We fix throughout:

- A prime cutoff $\kappa > 1$, with active prime set $S(\kappa) := \{p \text{ prime} : p \leq \kappa\}$ and $\pi(\kappa) := |S(\kappa)|$.
- The first N positive imaginary parts of nontrivial zeros of $\zeta(s)$, ordered $0 < \gamma_1 \leq \gamma_2 \leq \dots \leq \gamma_N$.
- A Gaussian damping parameter $\varepsilon > 0$.

Definition 2.1 (H_{str} and H_{null} , from Paper 3).

$$H_{\text{str}} := \ell^2(S(\kappa)), \quad \dim H_{\text{str}} = \pi(\kappa). \quad H_{\text{null}} := \ell^2(\{\gamma_k\}_{k=1}^N), \quad \dim H_{\text{null}} = N.$$

Both spaces carry standard inner products. We work throughout with real vectors and real operators.

Definition 2.2 (Resonance vectors, from Paper 3). *For each prime $p \in S(\kappa)$, the resonance vector $a_p \in H_{\text{null}}$ is defined by*

$$a_p := \left(e^{-\varepsilon^2 \gamma_k^2 / 2} \sin(\gamma_k \log p) \right)_{k=1}^N.$$

Since H_{null} is finite-dimensional ($\dim H_{\text{null}} = N$), $a_p \in H_{\text{null}}$ automatically; the Gaussian factor $e^{-\varepsilon^2 \gamma_k^2 / 2}$ is the damping used in the zero-coordinate representation.

Definition 2.3 (Linear map Φ , from Paper 3).

$$\Phi: H_{\text{str}} \rightarrow H_{\text{null}}, \quad \Phi(e_p) = a_p, \quad \Phi(c) = \sum_{p \leq \kappa} c_p a_p.$$

Definition 2.4 (Loop operator T , from Paper 3).

$$T := \Phi^* \circ \Phi: H_{\text{str}} \rightarrow H_{\text{str}}, \quad T_{pq} = \langle a_p, a_q \rangle_{H_{\text{null}}} = G_{pq}^{\text{un}}.$$

T is self-adjoint, positive semi-definite, and identical to the Gram matrix G^{un} of the resonance vectors.

Definition 2.5 (Canonical weight vector, from Paper 2). *The canonical weight vector $c \in H_{\text{str}}$*

used throughout this paper has components

$$c_p := \sqrt{f_p}, \quad f_p := \frac{4(\log p)^2(2p-1)}{p(p-1)^2},$$

the Weil-renormalised weight from Paper 2. It differs from Paper 3's η -framework weight $c_p^\eta = \sqrt{V_p(\frac{1}{2})}$; see Paper 3, Notation and Conventions and §4. The leading asymptotic for large p is $c_p \sim 2\sqrt{2} \frac{\log p}{\sqrt{p}} + O\left(\frac{\log p}{p^{3/2}}\right)$, since $f_p \sim 8(\log p)^2/p$. The exact formula is used throughout this paper and in all verification scripts. The local prime curvature $V_p(\sigma)$ of Paper 1 is related to f_p by $f_p = V_p(\frac{1}{2}) - 4(\log p)^2/p$ (subtraction of the leading divergent term).

Reference values for the Paper 4 weight $c_p = \sqrt{f_p}$: $c_p|_{p=2} \approx 1.698$, $c_p|_{p=3} \approx 1.418$, $c_p|_{p=5} \approx 1.080$, $c_p|_{p=7} \approx 0.884$.

Definition 2.6 (Energy quantities, from Paper 3).

$$E_{\text{str}} := \sum_{p \leq \kappa} c_p^2 \|a_p\|^2 = c^T D_a c, \quad E_{\text{spec}} := \left\| \sum_{p \leq \kappa} c_p a_p \right\|^2 = \|\Phi c\|^2,$$

where $D_a := \text{diag}(\|a_p\|^2)_{p \in S(\kappa)}$. The energy asymmetry is

$$\Delta := E_{\text{str}} - E_{\text{spec}} \geq 0 \quad (\text{numerically}), \quad \eta_{\text{ren}} := \frac{\Delta}{E_{\text{str}}} = 1 - \frac{c^T G^{\text{un}} c}{E_{\text{str}}} = \eta[c^{\text{ren}}].$$

Here $\eta[c]$ is the asymmetry functional evaluated at weight c ; this paper uses $c = c^{\text{ren}} := (c_p^{\text{ren}})_{p \leq \kappa}$ with $c_p^{\text{ren}} := \sqrt{f_p}$.

2.2 Definition of the Tehrani Operator $\tilde{T}$

Definition 2.7 (Tehrani operator).

$$\tilde{T} := \Phi \circ \Phi^*: H_{\text{null}} \rightarrow H_{\text{null}}.$$

Matrix representation. In the standard basis $\{f_k\}_{k=1}^N$ of H_{null} :

$$\tilde{T}_{kl} = \sum_{p \leq \kappa} (a_p)_k (a_p)_l = \sum_{p \leq \kappa} e^{-\varepsilon^2 \gamma_k^2 / 2} \sin(\gamma_k \log p) \cdot e^{-\varepsilon^2 \gamma_l^2 / 2} \sin(\gamma_l \log p).$$

Proposition 2.8 (Properties of $\tilde{T}$; proved). The operator $\tilde{T}$ satisfies:

- (i) **Self-adjointness:** $\tilde{T}^* = (\Phi \Phi^*)^* = \Phi \Phi^* = \tilde{T}$.
- (ii) **Positive semi-definiteness:** For all $u \in H_{\text{null}}$, $\langle \tilde{T}u, u \rangle = \|\Phi^* u\|^2 \geq 0$.
- (iii) **Rank bound:** $\text{rank}(\tilde{T}) \leq \min\{N, \pi(\kappa)\}$.

Proof. Parts (i) and (ii) are immediate from $\tilde{T} = \Phi \Phi^*$ using standard properties of adjoints. Part (iii) follows from $\text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B))$ applied to $\tilde{T} = \Phi \Phi^*$, giving

$\text{rank}(\tilde{T}) \leq \min(N, \pi(\kappa))$ (the bound from the singular values of $\Phi \in \mathbb{R}^{N \times m}$). The reverse inequality $\text{rank}(\tilde{T}) \geq \pi(\kappa)$ would require linear independence of $\{a_p\}_{p \leq \kappa}$ in H_{null} ; this is not proved analytically and is listed in §6. $\square$

Remark (Non-circularity of $\tilde{T}$). $\tilde{T}$ is built directly from prime resonance vectors $\{a_p\}$. The zero ordinates γ_k are numerical inputs to the resonance vectors and hence to $\tilde{T}$. Non-circularity means: no RH, GUE, or HP assumption is used; the γ_k are external inputs, not conclusions of the model.

Numerical evidence. At benchmark parameters $(\kappa, \varepsilon, N) = (53, 0.05, 100)$ the spectrum of $\tilde{T}$ exhibits a sharp gap between the $\pi(\kappa) = 16$ largest eigenvalues ($\mu_{16} \approx 4.94 \cdot 10^{-6}$) and the remaining ones ($\mu_{17} \approx 1.81 \cdot 10^{-18}$), giving the ratio $\mu_{17}/\mu_{16} \approx 3.7 \cdot 10^{-13}$. The numerical rank therefore equals $\pi(\kappa)$ at machine precision, consistent with linear independence of $\{a_p\}$ and hence with $\text{rank}(\tilde{T}) = \pi(\kappa)$, but the analytic statement remains an open problem.

2.3 Algebraic Spectral Identity

Theorem 2.9 (Spectral identity; proved).

$$\sigma(T) \setminus \{0\} = \sigma(\tilde{T}) \setminus \{0\}.$$

Proof. Let Φ be represented by an $N \times m$ real matrix ($m = \pi(\kappa)$). The nonzero eigenvalues of $T = \Phi^* \Phi$ and $\tilde{T} = \Phi \Phi^*$, counted with algebraic multiplicity, are the squared nonzero singular values of Φ . Hence $\sigma(T) \setminus \{0\} = \sigma(\tilde{T}) \setminus \{0\}$. The zero-eigenvalue multiplicities differ by the dimension gap: if $N \geq m$, then $\tilde{T}$ has $N - m$ additional zero eigenvalues; if $m > N$, then T has $m - N$ additional zero eigenvalues. At the benchmark $(\kappa, N) = (53, 100)$, this gives $N - m = 100 - 16 = 84$ additional zero eigenvalues for $\tilde{T}$. $\square$

Remark (Numerical implementation check). At benchmark parameters $(\kappa, \varepsilon, N) = (53, 0.05, 100)$:

$$\max_{j=1}^{\pi(\kappa)} |\lambda_j(T) - \mu_j(\tilde{T})| < 10^{-15} \quad (\text{machine precision}).$$

This verifies the algebraic identity numerically.

Remark. The identity $\lambda_j(T) = \mu_j(\tilde{T})$ is an algebraic theorem — it holds in exact arithmetic for any choice of κ , ε , and N . The eigenvalues $\lambda_j = \mu_j$ are not the zero ordinates γ_k ; they are constructive spectral quantities arising from Φ .

2.4 $\tilde{T}$ is NOT a Hilbert–Pólya Operator

This section records a negative result.

Proposition 2.10 (Hilbert–Pólya diagnostic failure; numerical). *On the tested finite-cutoff grid, $\tilde{T}$ and $W_1 = C_T \tilde{T}^+$ give no numerical support for an asymptotic Hilbert–Pólya interpretation:*

- (a) [Numerical] *The eigenvalues of $\tilde{T}$ satisfy $\mu_j \approx C_T(\kappa)/\gamma_{k(j)}$ (finite-grid OLS fit) — they are inversely proportional to zero ordinates (§ 3.1).*
- (b) [Numerical] *The zero ordinates γ_k appear as localization centers of eigenvectors v_j (§ 3.2), not as eigenvalues.*
- (c) [Numerical] *The inverse operator $W_1 := C_T \cdot \tilde{T}^+$ has eigenvalues $\omega_j = C_T/\mu_j \approx \gamma_{k(j)}$ for small j , but the correlation $r_2 := \text{corr}(\omega_j, \gamma_{k(j)})$ falls from 0.502 (at $\kappa = 23$) to 0.160 (at $\kappa = 503$). On the tested grid, W_1 fails the HP-correlation diagnostic; these data give no numerical support for W_1 as an asymptotic Hilbert–Pólya operator.*

The fundamental distinction:

$$\text{HP-operator: eigenvalues} = \gamma_k. \quad \tilde{T}: \text{eigenvector localization centers} = \gamma_k.$$

$\tilde{T}$ is a **resonance operator**: it encodes the γ_k as mode centers, not as spectral values.

Proof of (a)(b)(c). Numerical; see § 3.1, § 3.2, § 3.4 for detailed data. $\square$

Remark (Numerical obstruction pattern). *Gaussian damping $e^{-\varepsilon^2 \gamma_k^2}$ is a plausible explanation for the finite-grid failure of the HP-correlation diagnostic. For $j = 16$ at $\kappa = 53$, the dominant zero is γ_1 , and the HP-correlation r_2 falls to 0.16 on the tested grid. This finite-grid pattern is consistent with the observed failure of W_1 as an asymptotic HP-candidate; no structural obstruction theorem is proved here.*

3 Spectral Analysis of $\tilde{T}$

3.1 Eigenvalue Structure: $\mu_j \approx C_T(\kappa)/\gamma_{k(j)}$ (finite-grid OLS)

Numerical evidence (Eigenvalue–Zero Ordinate Correlation).

Let $k_{\text{dom}}(j) := \arg \max_k |(v_j)_k|$ be the dominant zero index for eigenvector v_j . Then:

j	$\mu_j(\tilde{T})$	k_{dom}	$\gamma_{k_{\text{dom}}}$	Localization
1	4.219803	1	14.1347	0.985
2	2.565694	3	25.0109	0.741
3	1.744649	2	21.0220	0.745
4	0.986492	4	30.4249	0.854
5	0.314199	5	32.9351	0.794
6	0.217900	6	37.5862	0.885

The OLS correlation between μ_j and $1/\gamma_{k(j)}$ is

$$r_1 = \text{corr}(\mu_j, 1/\gamma_{k(j)}) = 0.950 \quad (\kappa = 53).$$

For $\kappa = 503$: $r_1 = 0.859$ (weakening as κ grows, consistent with Gaussian damping).

Remark. *The dominant index satisfies $k_{\text{dom}}(1) = 1$ for all tested $\kappa \in \{23, 53, 101, 199, 503, 1009\}$: the first eigenvector of $\tilde{T}$ is always associated with $\gamma_1 = 14.1347$.*

Remark (Cauchy–Schwarz warning). *Note that $r_1 = 0.950$ coexists with $r_2 = 0.502$ (§ 3.4). These measure different things: r_1 tests monotone ordering; r_2 tests value-level approximation. The distinction between ordering and value accuracy is crucial for the HP assessment in § 2.4.*

3.2 Eigenvector Localization

Numerical evidence (Eigenvector Localization).

For each eigenvector v_j of $\tilde{T}$ (normalized: $\|v_j\| = 1$), the localization is measured by $L_j := |(v_j)_{k_{\text{dom}}(j)}|$. Observed range: $L_j \in [0.45, 0.99]$ for $j = 1, \dots, 14$.

For comparison, a uniformly random unit vector in $\mathbb{R}^{100}$ has typical entry magnitude $1/\sqrt{100} = 0.10$. The observed localization is 4.5–9.9 times the random baseline. The localization structure suggests that $\tilde{T}$ encodes the ordering of zero ordinates in a distributional sense: each mode v_j is concentrated near the zero $\gamma_{k(j)}$ in the zero-space H_{null} . This is the precise content of calling $\tilde{T}$ a resonance operator.

Assumption 3.1 (Nonzero resonance norms). *For the finite parameter sets under consideration, $\|a_p\|_{H_{\text{null}}} > 0$ for every active prime $p \leq \kappa$. Equivalently, D_a is positive diagonal, and $D_a^{-1/2}$ is well-defined. This is verified numerically for all parameter sets used below.*

3.3 The Two Arithmetic Constants C_η and C_T

Definition 3.2 (Arithmetic constants).

$$R_\eta(\kappa) := \frac{\lambda_{\max}(T_{\text{ren}})}{\pi(\kappa)}, \quad C_\eta \approx 0.39 \quad (\text{observed finite-grid value; no limit theorem}),$$

where $T_{\text{ren}} := D_a^{-1/2} T D_a^{-1/2}$.

$$C_T(\kappa) := \text{OLS-slope of } \mu_j \sim C_T / \gamma_{k(j)}, \quad C_T / \pi(\kappa) \in \{2.49, 2.21, 2.17, 1.94, 1.89\}$$

for $\kappa \in \{23, 53, 101, 199, 503\}$ (weakly decreasing).

$$C_\eta \neq C_T.$$

These are two distinct arithmetic quantities; always use subscripts.

More explicitly, let $J(\kappa) = \{1, \dots, r(\kappa)\}$ where $r(\kappa) = \text{rank}_{\text{num}}(\tilde{T})$, and let $k_{\text{dom}}(j) = \arg \max_k |(v_j)_k|$. Then:

$$C_T(\kappa) := \arg \min_{C \in \mathbb{R}} \sum_{j \in J(\kappa)} \left(\mu_j - \frac{C}{\gamma_{k_{\text{dom}}(j)}} \right)^2.$$

No intercept term is used.

The ratio $R_\eta(\kappa)$ is numerically stable at $C_\eta \approx 0.39$ across $\kappa \in \{23, \dots, 1009\}$ (see table; $\sigma^2 < 0.0003$). Whether $R_\eta(\kappa)$ converges to a limit is analytically open; we write C_η as a name for the observed value.

Numerical evidence ($\lambda_{\max, \text{ren}}$ growth).

κ	$\pi(\kappa)$	$\lambda_{\max, \text{ren}}$	Ratio
23	9	3.59	0.399
53	16	6.38	0.399
101	26	10.35	0.398
199	46	17.76	0.386
503	96	37.38	0.389
1009	169	65.16	0.386

Variance of the ratio: $\sigma^2 < 0.0003$ — extremely stable. The universal bound $\lambda_{\max, \text{ren}} < 1$ (an earlier conjecture in this series, now falsified) is numerically falsified: $\lambda_{\max, \text{ren}} > 1$ already on the tested grid. The data are compatible with growth proportional to $\pi(\kappa)$, with observed ratio $R_\eta(\kappa) \approx 0.39$ (see Definition 3.2); convergence of this ratio remains analytically open.

Remark (Analytical status of C_η). *The observed ratio $R_\eta(\kappa) \approx 0.39$ (denoted C_η) is not analytically explained. A candidate formula $C_\eta = 1/I_N(\varepsilon)$ is **falsified**: it gives $1/I_{100}(0.05) \approx 0.73 \neq 0.39$. The correct representation involves a harmonic mean (analytically open; see Open Problem 6.5).*

3.4 The Inverse Operator $W_1 = C_T \cdot \tilde{T}^+$

Definition 3.3 (Inverse operator). $W_1 := C_T \cdot \tilde{T}^+$, where $\tilde{T}^+$ denotes the Moore–Penrose pseudoinverse of $\tilde{T}$ and C_T is the OLS constant from Definition 3.2.

Proposition 3.4 (Self-adjointness of W_1 ; proved). W_1 is self-adjoint.

Proof. $\tilde{T}$ is self-adjoint (Proposition 2.8(i)). For any self-adjoint operator, the Moore–Penrose pseudoinverse is also self-adjoint (standard result in linear algebra). C_T is a real scalar. Therefore $W_1 = C_T \cdot \tilde{T}^+$ is self-adjoint. $\square$

Numerically: $\|W_1 - W_1^T\|_\infty = 3.1 \times 10^{-4}$, $\|W_1 - W_1^T\|_\infty / \|W_1\|_\infty \approx 6.7 \times 10^{-11}$ (rounding from pseudoinversion of a moderately conditioned matrix).

Proposition 3.5 (Non-circularity status of W_1 ; proved). $W_1 = C_T \cdot \tilde{T}^+$ does not assume RH or the Hilbert–Pólya postulate. However, its normalization constant C_T is the OLS slope in $\mu_j \approx C_T(\kappa)/\gamma_{k(j)}$ (finite-grid OLS fit), which is calibrated against the zero ordinates. W_1 should therefore not be presented as fully input-free. $\tilde{T}$ is explicit and deterministic; the γ_k are external numerical inputs, not conclusions of the model. Proof. $\tilde{T}$ is constructed from prime resonance vectors (Definition 2.7), using γ_k as numerical inputs. $\tilde{T}^+$ is the Moore–Penrose pseudoinverse

— *no postulate. The scalar C_T is computed via OLS regression of μ_j against $1/\gamma_{k(j)}$: this uses γ_k as calibration values, but does not assume RH or HP.* $\square$

Numerical evidence (W_1 HP-correlation failure).

κ	$\pi(\kappa)$	$r_2 = \text{corr}(\omega_j, \gamma_{k(j)})$
23	9	0.502
53	16	0.436
101	26	0.321
199	46	0.226
503	96	0.160

The decreasing trend indicates, on the tested grid, that W_1 is not supported as an asymptotic HP-operator. The observed pattern is consistent with Gaussian damping: $e^{-\varepsilon^2 \gamma_k^2/2}$ in Φ is a plausible explanation for the rapid decay of μ_j , with a fitted power-law exponent $\alpha \approx -4.88$. This decay is much steeper than the approximate $1/\gamma$ behaviour that would be needed for the pseudoinverse eigenvalues $\omega_j = C_T/\mu_j$ to track the zero ordinates. Pseudoinversion therefore strongly amplifies tail modes.

Summary for § 3.4: W_1 is self-adjoint (proved). Its construction does not assume RH or the HP postulate, but C_T is calibrated via OLS on the zero ordinates — W_1 is therefore not fully input-free (Proposition 3.5). W_1 is the natural HP-candidate but fails numerically as κ grows. These diagnostics give no numerical support for this particular pseudoinverse construction as an asymptotic Hilbert–Pólya candidate. This does not preclude other spectral functions $f(\tilde{T})$ from being valid HP-candidates; see § 6.

4 Monotonicity and Prime Exponential Sums

The following estimates record ingredients for a possible analytic route toward bounding η_{ren} from below, based on the interplay of three structural features: the monotone weight sequence (c_p) , the oscillatory mode structure $(\sin(\gamma_k \log p))$, and Abel summation.

4.1 Monotone weight structure

Lemma 4.1 (Monotone weight structure; proved for $p \geq 11$). *The canonical weight $c_p = \sqrt{f_p}$ is strictly monotone decreasing in p for all primes $p \geq 11$.*

Proof. *The exact expression for c_p is*

$$c_p = \frac{2 \log p \sqrt{2p-1}}{\sqrt{p}(p-1)}.$$

Taking the logarithmic derivative:

$$\frac{d}{dp} \log c_p = \frac{1}{p \log p} + \frac{1}{2p-1} - \frac{1}{2p} - \frac{1}{p-1}.$$

For $p \geq 11$, we have $\log p > 2$, hence $1/(p \log p) < 1/(2p)$. Therefore:

$$\frac{d}{dp} \log c_p < \frac{1}{2p} + \frac{1}{2p-1} - \frac{1}{2p} - \frac{1}{p-1} = \frac{1}{2p-1} - \frac{1}{p-1} < 0,$$

since $2p-1 > p-1 > 0$. Hence c_p is strictly decreasing in p for all $p \geq 11$. $\square$

4.2 Oscillatory mode structure

Lemma 4.2 (Oscillatory mode intervals; proved). *For fixed $\gamma_k > 0$, the function $p \mapsto \sin(\gamma_k \log p)$ has zeros at $p = e^{m\pi/\gamma_k}$ for integers $m \geq 0$, and its sign is constant on each open interval $(e^{m\pi/\gamma_k}, e^{(m+1)\pi/\gamma_k})$, alternating between adjacent intervals. Restricting to primes $p \geq 2$ selects the relevant subcollection.*

Proof. Direct from $\sin(x) = 0$ at $x = m\pi$: the argument $\gamma_k \log p$ crosses $m\pi$ at $p = e^{m\pi/\gamma_k}$. $\square$

Remark (Prime occupation; open). *For fixed γ_k , the PNT suggests eventual prime occupation of the intervals I_m as $\kappa \rightarrow \infty$; a quantitative finite-cutoff version with the required uniformity is not proved here.*

4.3 Abel Summation Principle

Lemma 4.3 (Abel summation principle; proved). *Let $P \geq 1$. Let $(f_1, \dots, f_P)$ be monotone non-increasing and non-negative. Let $(g_1, \dots, g_P)$ be any sequence with partial sums bounded by M : $|G_m| := |\sum_{i=1}^m g_i| \leq M$ for all m . Then:*

$$\left| \sum_{i=1}^P f_i g_i \right| \leq M \cdot f_1.$$

Proof. Apply the Abel partial summation formula: $\sum_{i=1}^P f_i g_i = f_P G_P - \sum_{i=1}^{P-1} (f_{i+1} - f_i) G_i$. Taking absolute values:

$$\begin{aligned} \left| \sum_{i=1}^P f_i g_i \right| &\leq |f_P| \cdot |G_P| + \sum_{i=1}^{P-1} |f_{i+1} - f_i| \cdot |G_i| \\ &\leq M f_P + M \sum_{i=1}^{P-1} (f_i - f_{i+1}) \quad [f \text{ decreasing}] \\ &= M f_P + M(f_1 - f_P) = M f_1. \quad \square \end{aligned}$$

Remark (Application context for Paper 4). *Lemma 4.1 establishes monotonicity of the unnormalised weights c_{p_i} . In the Rayleigh quotient identity of § 5.1, the relevant normalised vector is $\tilde{c} = D_a^{1/2} c / \sqrt{E_{\text{str}}}$, whose components satisfy $(\tilde{c})_p \propto c_p \|a_p\|$. Monotonicity of c_p therefore does*

not by itself imply monotonicity of the weights entering the spectral projections, since $\|a_p\|$ depends on the zero-sum profile and may disturb the ordering. Applying Lemma 4.3 to spectral projections therefore requires either a separate monotonicity or bounded-variation estimate for $c_p\|a_p\|$, or a different transfer lemma from prime-phasor partial sums to the normalised spectral projections $|\langle u_j, \tilde{c} \rangle|$. This is the missing transfer step noted in Open Problem 6.3.

4.4 Prime Exponential Endpoint Estimate

Proposition 4.4 (Prime exponential endpoint estimate; proved). *Let $t \neq 0$ be fixed. As $x \rightarrow \infty$:*

$$\sum_{p \leq x} p^{it} = \frac{x^{1+it}}{(1+it)\log x} + O_t\left(\frac{x}{\log^2 x}\right).$$

In particular, for the sine sum:

$$M_k(x) := \max_{y \leq x} \left| \sum_{p \leq y} \sin(\gamma_k \log p) \right| \leq \left(\frac{1}{\sqrt{1+\gamma_k^2}} + o_{\gamma_k}(1) \right) \pi(x).$$

Proof. Write $\vartheta(x) = \sum_{p \leq x} \log p$ (Chebyshev's function). By Abel partial summation, $\sum_{p \leq x} p^{it} = \int_2^x u^{it} \frac{d\vartheta(u)}{\log u}$. The PNT gives $\vartheta(x) = x + o(x)$, and with a quantitative error term $\vartheta(x) = x + O(xe^{-c\sqrt{\log x}})$ (Hadamard/de la Vallée-Poussin); see [11], § 6. Writing $E(u) = \vartheta(u) - u$, Stieltjes integration by parts gives

$$\int_2^x F(u) dE(u) = F(x)E(x) - F(2)E(2) - \int_2^x E(u)F'(u) du,$$

where $F(u) = u^{it}/\log u$ and $F'(u) = (it \log u - 1)u^{it-1}/(\log u)^2 = O_t(1/(u \log u))$. The de la Vallée-Poussin bound $E(u) = O(ue^{-c\sqrt{\log u}})$ implies $E(x)F(x) = O_t(xe^{-c\sqrt{\log x}})$ and $\int_2^x |E(u)F'(u)| du = O_t\left(\int_2^x e^{-c\sqrt{\log u}} du\right) = O_t(xe^{-c'\sqrt{\log x}})$. Hence $\int_2^x F(u) dE(u) = O_t(x/\log^2 x)$, and the main term comes from $\int_2^x u^{it}/\log u du$, yielding the displayed asymptotic. Integrating by parts yields the first display:

$$\int_2^x \frac{u^{it}}{\log u} du = \frac{x^{1+it}}{(1+it)\log x} + O_t\left(\frac{x}{\log^2 x}\right),$$

where the de la Vallée-Poussin error is absorbed into the same O_t -term for fixed t . The estimate holds for fixed t ; no uniformity in t or zero index k is asserted. The sine sum bound follows by taking imaginary parts:

$$\left| \sum_{p \leq x} \sin(t \log p) \right| = \left| \operatorname{Im} \sum_{p \leq x} p^{it} \right| \leq \frac{x}{(1+t^2)^{1/2} \log x} + O_t\left(\frac{x}{\log^2 x}\right) = \left(\frac{1}{\sqrt{1+t^2}} + o_t(1) \right) \pi(x). \quad \square$$

The same estimate applies with y in place of x for any y in the range. For fixed t , the constant in the $O_t(\cdot)$ -term is uniform for all $y \geq y_0(t)$. Since $y/\log y$ and $y/\log^2 y$ are increasing beyond a fixed threshold, the same bound holds after taking the supremum over $y_0(t) \leq y \leq x$. The finitely many values $2 \leq y < y_0(t)$ contribute $O_t(1)$, hence $o_t(\pi(x))$ as $x \rightarrow \infty$; they do not affect the

maximum asymptotic.

Remark (Methodological note). *The estimate, for each fixed γ_k , $M_k(\kappa) \leq (1/\sqrt{1+\gamma_k^2} + o_{\gamma_k}(1))\pi(\kappa)$ (no uniformity in k) is a standard consequence of the Prime Number Theorem; the analytic method has precedents in Turán’s program on approximative Dirichlet polynomials [16]. What is specific to this paper is how this bound may enter a future analytic route to the energy-asymmetry problem.*

Remark (What Proposition 4.4 does NOT prove). *Proposition 4.4 does not prove $M_k(\kappa) = o(\pi(\kappa))$ for fixed k as $\kappa \rightarrow \infty$. This stronger statement is related to the pair-cancellation hypothesis (E_{pair}) stated in § 5.3.*

5 Results on $\eta_{\text{ren}} > 0$

5.1 Numerical Confirmation

Numerical evidence ($\eta_{\text{ren}} > 0$ on the tested grid $\kappa \in \{23, 53, 101, 199, 503, 1009\}$).

Let $\tilde{c} = D_a^{1/2}c/\sqrt{E_{\text{str}}}$ be the normalized canonical weight vector. The Rayleigh quotient identity

$$\langle T_{\text{ren}} \tilde{c}, \tilde{c} \rangle = 1 - \eta_{\text{ren}}$$

holds to machine precision ($< 2 \times 10^{-16}$), where $T_{\text{ren}} := D_a^{-1/2} T D_a^{-1/2}$.

κ	$\pi(\kappa)$	$\langle T_{\text{ren}} \tilde{c}, \tilde{c} \rangle$	$< 1?$	η_{ren}
23	9	0.18849	✓	0.81151
53	16	0.33073	✓	0.66927
101	26	0.22803	✓	0.77197
199	46	0.21740	✓	0.78260
503	96	0.21788	✓	0.78212
1009	169	0.18002	✓	0.81998

$\eta_{\text{ren}} > 0$ on the tested grid $\kappa \in \{23, 53, 101, 199, 503, 1009\}$ ✓.

Collective mechanism. Writing $T_{\text{ren}} u_j = \lambda_j u_j$, define the mode contribution

$$q_j := \lambda_j |\langle u_j, \tilde{c} \rangle|^2,$$

so that $\langle T_{\text{ren}} \tilde{c}, \tilde{c} \rangle = \sum_j q_j$. At $\kappa = 53$: $q_4 = 0.193$, $q_1 = 0.040$. The dominant contribution comes from mode $j = 4$, not from the dominant eigenmode $j = 1$. Approximately 92% cancellation occurs at $j = 1$ ($\kappa = 53$). The effect is distributed across ~ 8 eigenmodes — $\eta_{\text{ren}} > 0$ is a **multi-mode Rayleigh-quotient contribution**, not a simple u_1 -orthogonality principle.

5.2 Δ -Decomposition and Burst Stability

Lemma 5.1 (Exact energy-gap decomposition; proved). *Let $S(r) := \sum_{k=1}^N e^{-\varepsilon^2 \gamma_k^2} \cos(\gamma_k \log r)$. Then*

$$\Delta = \Delta_{\text{Burst}} + \Delta_{\text{Cross}} + \Delta_{\text{Stream}},$$

where

$$\begin{aligned} \Delta_{\text{Burst}} &= \sum_{\substack{p < q \leq 7 \\ p, q \text{ prime}}} c_p c_q [S(pq) - S(p/q)], \\ \Delta_{\text{Cross}} &= \sum_{p \leq 7 < q \leq \kappa} c_p c_q [S(pq) - S(p/q)], \\ \Delta_{\text{Stream}} &= \sum_{7 < p < q \leq \kappa} c_p c_q [S(pq) - S(p/q)]. \end{aligned}$$

Proof. By definition,

$$E_{\text{spec}} = \left\| \sum_{p \leq \kappa} c_p a_p \right\|^2 = \sum_{p, q \leq \kappa} c_p c_q \langle a_p, a_q \rangle.$$

Since $E_{\text{str}} = \sum_{p \leq \kappa} c_p^2 \|a_p\|^2$, the diagonal terms ($p = q$) cancel in $\Delta = E_{\text{str}} - E_{\text{spec}}$, giving

$$\Delta = -2 \sum_{p < q \leq \kappa} c_p c_q \langle a_p, a_q \rangle.$$

Using $\cos(A + B) - \cos(A - B) = -2 \sin A \sin B$ with $A = \gamma_k \log p$, $B = \gamma_k \log q$:

$$-2 \langle a_p, a_q \rangle = \sum_{k=1}^N e^{-\varepsilon^2 \gamma_k^2} [\cos(\gamma_k \log(pq)) - \cos(\gamma_k \log(p/q))] = S(pq) - S(p/q).$$

Splitting the double sum by ranges of p, q gives the asserted decomposition. $\square$

$\square$

Numerical evidence (Δ_{Burst} stability).

Pair	$S(pq)$	$S(p/q)$	$c_p c_q$	Contribution
(2, 3)	0.9536	0.2653	≈ 2.408	+1.657
(2, 5)	0.4110	0.6555	≈ 1.833	-0.448
(2, 7)	0.5794	0.5812	≈ 1.501	-0.003
(3, 5)	0.9440	0.3686	≈ 1.531	+0.881
(3, 7)	0.7193	0.5571	≈ 1.254	+0.203
(5, 7)	0.9668	0.1135	≈ 0.954	+0.814
Total				$\approx +\mathbf{3.11}$

At fixed (ε, N) , Δ_{Burst} is independent of κ for all $\kappa \geq 7$ by definition (the sum is over the fixed pairs $p < q \leq 7$). Numerically, at $(\varepsilon, N) = (0.05, 100)$: $\Delta_{\text{Burst}} = 3.105 \dots > 0$ (Lemma 5.3).

The sufficient condition for $\Delta > 0$ is:

$$|\Delta_{\text{Cross}} + \Delta_{\text{Stream}}| < \Delta_{\text{Burst}} = 3.105.$$

This is numerically satisfied on the tested grid $\kappa \in \{23, 53, 101, 199, 503, 1009\}$; analytic proof beyond this grid is OPEN.

5.3 Conditional Result under (E_{rem})

Assumption 5.2 ((E_{rem}) ; remainder control). *At fixed reference parameters $(\varepsilon, N) = (0.05, 100)$ and with the Paper-4 weight $c_p^{\text{ren}} = \sqrt{f_p}$, there exist $\rho < 1$ and $\kappa_0 \geq 7$ such that for all $\kappa \geq \kappa_0$:*

$$|\Delta_{\text{Cross}}(\kappa) + \Delta_{\text{Stream}}(\kappa)| \leq \rho \cdot \Delta_{\text{Burst}}.$$

Lemma 5.3 (Burst positivity; numerical). *At benchmark parameters $(\varepsilon, N) = (0.05, 100)$, the finite six-pair burst sum satisfies*

$$\Delta_{\text{Burst}} = \sum_{\substack{p < q \leq 7 \\ p, q \text{ prime}}} c_p^{\text{ren}} c_q^{\text{ren}} (S(pq) - S(p/q)) = 3.105 \in [3.09, 3.12].$$

In particular $\Delta_{\text{Burst}} > 0$. Numerical verification code is available at <https://github.com/utehrani/analysislab-nt>.

Theorem 5.4 (Conditional/numerical positivity from remainder control). *At the reference parameters $(\sigma_0, \varepsilon, N) = (\frac{1}{2}, 0.05, 100)$, under the numerical bound $\Delta_{\text{Burst}} > 0$ (Lemma 5.3), the exact decomposition of Lemma 5.1, Assumption (E_{rem}) , and $E_{\text{str}}(\kappa) > 0$ (Assumption 3.1 and $c_p^{\text{ren}} > 0$), for all $\kappa \geq \kappa_0$:*

$$\Delta(\kappa) \geq (1 - \rho) \Delta_{\text{Burst}} > 0,$$

and consequently

$$\eta_{\text{ren}}(\kappa) = \frac{\Delta(\kappa)}{E_{\text{str}}(\kappa)} \geq \frac{(1 - \rho) \Delta_{\text{Burst}}}{E_{\text{str}}(\kappa)} > 0.$$

Proof. By the triangle inequality and (E_{rem}) :

$$\Delta(\kappa) = \Delta_{\text{Burst}} + \Delta_{\text{Cross}}(\kappa) + \Delta_{\text{Stream}}(\kappa) \geq \Delta_{\text{Burst}} - |\Delta_{\text{Cross}}(\kappa) + \Delta_{\text{Stream}}(\kappa)| \geq (1 - \rho) \Delta_{\text{Burst}} > 0.$$

By Assumption 3.1 and $c_p^{\text{ren}} > 0$, $E_{\text{str}}(\kappa) > 0$; hence division by $E_{\text{str}}(\kappa)$ gives $\eta_{\text{ren}}(\kappa) \geq (1 - \rho) \Delta_{\text{Burst}} / E_{\text{str}}(\kappa) > 0$. $\square$

Remark (Numerical verification of (E_{rem})). *On the tested grid $\kappa \in \{23, 53, 101, 199, 503, 1009\}$ with $\varepsilon = 0.05$, $N = 100$:*

κ	$\Delta(\kappa)$	Δ_{Burst}	$ \Delta_{\text{Cross}} + \Delta_{\text{Stream}} $	$\rho(\kappa)$
23	4.205	3.105	1.100	0.354
53	3.707	3.105	0.601	0.194
101	4.437	3.105	1.332	0.429
199	4.592	3.105	1.486	0.479
503	4.662	3.105	1.556	0.501
1009	4.915	3.105	1.809	0.583

The maximum observed ratio is $\rho_{\max} = 0.583 < 1$, consistent with (E_{rem}) for any $\rho \in (0.583, 1)$. Whether $\rho(\kappa)$ remains below 1 for all κ is analytically OPEN.

Remark (Route toward (E_{rem})). A stronger but more structured hypothesis (E_{osc}) would assert uniform cancellation of weighted prime-phasor partial sums:

$$\left| \sum_{p \leq \kappa} c_p^2 \sin(\gamma_{k_1} \log p) \sin(\gamma_{k_2} \log p) \right| \leq C \cdot \frac{\Delta_{\text{Burst}}}{\pi(\kappa)^A}$$

for some $C > 0$, $A \geq 1$, independent of k_1, k_2 . Such an estimate would still have to be combined with a separate aggregation lemma controlling the passage from these pairwise oscillatory sums to $|\Delta_{\text{Cross}} + \Delta_{\text{Stream}}|$. That implication is not proved here and is part of the open problem; see Open Problem 6.2. The related hypothesis (E_{pair}) controls pairwise cancellation at a weaker level; its implication for (E_{rem}) is also not established.

Remark (Correction to earlier formulation). The earlier claim “ $\Delta(\kappa) \rightarrow \Delta_{\text{Burst}}$ ” was incorrect: $\Delta_{\text{Burst}} \approx 3.11$ is a κ -invariant positive contribution, not the limit. Numerically, $\Delta_{\infty} \approx 4.9 > \Delta_{\text{Burst}}$. It is observed to be a lower bound for $\Delta(\kappa)$ on the tested grid, but this lower-bound property is not proved without additional assumptions.

Remark (Numerical extrapolation). On the tested grid $\kappa \in \{23, 53, 101, 199, 503, 1009\}$, $\eta_{\text{ren}}(\kappa)$ remains positive and lies in $[0.669, 0.820]$. The data are compatible with a positive limiting value, but no convergence theorem is proved.

Remark (Connection to RH). Under (E_{rem}) , Theorem 5.4 gives $\eta_{\text{ren}} > 0$ at the stated reference parameters. A possible future bridge toward RH would require an independent transfer theorem connecting an appropriate limiting form of $\eta_{\text{ren}} > 0$ to zero exclusion off the critical line. No such limiting theorem or transfer principle is established here.

6 Open Problems

Open Problem 6.1 (Rank equality $\text{rank}(\tilde{T}) = \pi(\kappa)$). At benchmark parameters $(\kappa, \varepsilon, N) = (53, 0.05, 100)$, prove analytically that $\{a_p\}_{p \leq \kappa} \subset H_{\text{null}}$ are linearly independent — equivalently that $\text{rank}(\tilde{T}) = \pi(\kappa)$ rather than the unconditional bound $\text{rank}(\tilde{T}) \leq \pi(\kappa)$ established in Proposition 2.8 via the at-most inequality. Numerically confirmed: the spectral gap $\mu_{\pi(\kappa)+1}/\mu_{\pi(\kappa)} \approx 3.7 \cdot 10^{-13}$ at benchmark parameters is consistent with linear independence and hence with $\text{rank}(\tilde{T}) = \pi(\kappa)$ at machine precision.

Open Problem 6.2 (Proof of (E_{rem})). *Prove Assumption (E_{rem}) (Definition 5.2) at fixed reference parameters $(\varepsilon, N) = (0.05, 100)$: that there exist $\rho < 1$ and κ_0 such that $|\Delta_{\text{Cross}}(\kappa) + \Delta_{\text{Stream}}(\kappa)| \leq \rho \cdot \Delta_{\text{Burst}}$ for all $\kappa \geq \kappa_0$. Numerically $\rho_{\text{max}} = 0.583$ on the tested grid $\kappa \in \{23, 53, 101, 199, 503, 1009\}$; whether this remains below 1 for all κ is open.*

A structured approach via (E_{osc}) : prove

$$\left| \sum_{p \leq \kappa} c_p^2 \sin(\gamma_{k_1} \log p) \sin(\gamma_{k_2} \log p) \right| \leq C \cdot \frac{\Delta_{\text{Burst}}}{\pi(\kappa)^A}$$

*for some $C > 0$, $A \geq 1$. By itself, this estimate would **not** yet prove (E_{rem}) ; one would also need an aggregation lemma transferring the pairwise oscillatory control to $|\Delta_{\text{Cross}} + \Delta_{\text{Stream}}|$. The formulation and proof of such an aggregation lemma are part of this open problem. This approach is adjacent to classical questions on zero-ordinate independence and prime phase distribution, but it is a separate quantitative prime-phase cancellation problem. No implication to or from the classical Linear Independence conjecture is asserted here; for context see [10, 12, 9]. The related hypothesis (E_{pair}) controls pairwise cancellation at a weaker level; its implication for (E_{rem}) is also not established.*

Open Problem 6.3 (Analytic positivity of η_{ren}). *Prove $\eta_{\text{ren}}(\kappa) > 0$ for all $\kappa \geq 2$ without assuming (E_{rem}) or any unverified hypothesis (at fixed reference parameters $(\varepsilon, N) = (0.05, 100)$).*

Current best: Proposition 4.4 gives a PNT-based bound for raw prime exponential sums $\sum_{p \leq x} \sin(\gamma_k \log p)$ at fixed γ_k . It does not yet bound the spectral projections $|\langle u_j, \tilde{c} \rangle|$, where $\tilde{c} = D_a^{1/2} c / \sqrt{E_{\text{str}}}$ is the normalised vector in the Rayleigh quotient identity of § 5.1. A missing transfer lemma would be needed to pass from the prime-phaser bounds to the normalised spectral energy control

$$\sum_j \lambda_j |\langle u_j, \tilde{c} \rangle|^2 < 1.$$

Equivalently, in unnormalised form:

$$\sum_j \lambda_j |\langle u_j, D_a^{1/2} c \rangle|^2 < E_{\text{str}}.$$

Open Problem 6.4 (Circularity-free spectral function). *Two separate questions.*

Here “circularity-free” means that the functional calculus rule is fixed independently of any OLS fit, postulated HP spectrum, or post hoc matching to the target ordinates; the ordinates entering $\tilde{T}$ through Φ remain external inputs.

(a) Spectral function of the fixed $\tilde{T}$: *For fixed N , does any circularity-free function $f(\tilde{T}_{\kappa, N})$ have its first N spectral values converge, after a specified matching, to $\{\gamma_1, \dots, \gamma_N\}$ as $\kappa \rightarrow \infty$? The natural candidate $W_1 = C_T \cdot \tilde{T}^+$ fails: r_2 falls to 0.16 on the tested grid (§ 3.4). The observed obstruction is consistent with Gaussian damping; no structural obstruction theorem is proved here.*

(b) Model-variant question (separate from (a)): *If the Gaussian damping $e^{-\varepsilon^2 \gamma_k^2 / 2}$ in Φ were*

replaced by a slower decay $e^{-\varepsilon\gamma_k}$, could the resulting modified operator $\tilde{T}'$ support a circularity-free spectral function? This would define a different operator family and is not a spectral function of the present $\tilde{T}$.

Open Problem 6.5 (Arithmetic constant C_η). Determine the analytic formula for the observed ratio $R_\eta(\kappa) = \lambda_{\max, \text{ren}}/\pi(\kappa) \approx 0.39$ (denoted C_η ; see Definition 3.2). The candidate $C_\eta = 1/I_N(\varepsilon)$ is falsified. The correct representation involves a harmonic mean (analytically open).

Open Problem 6.6 (Asymptotics of C_T). Does $C_T(\kappa)/\pi(\kappa)$ converge as $\kappa \rightarrow \infty$? Numerically: values 2.49, 2.21, 2.17, 1.94, 1.89 (decreasing) — the limit, if it exists, is analytically undetermined.

Open Problem 6.7 (Monotone bound from prime exponential-sum control). Does $\eta_{\text{ren}} > 0$ follow from PNT-based prime exponential-sum control, with the zero ordinates γ_k treated as fixed external parameters (at fixed $(\varepsilon, N) = (0.05, 100)$)?

Evidence: $\Delta_{\text{Burst}} \approx 3.11 > 0$ depends only on primes $p \leq 7$ and is κ -invariant. Analytically OPEN.

Remark (Connection to RH). Under (E_{rem}) , Theorem 5.4 gives $\eta_{\text{ren}} > 0$. A possible future bridge toward RH would require an independent transfer theorem connecting an appropriate limiting form of $\eta_{\text{ren}} > 0$ to zero exclusion off the critical line. No such limiting theorem or transfer principle is established here.

Non-Circularity

We document explicitly that the results of this paper do not rely on the objects they are directed toward. The following holds unconditionally throughout §§ 2–5.

No RH as input. At no point is the Riemann Hypothesis assumed. The zero ordinates γ_k enter the construction of Φ as numerically specified inputs, and no property of their distribution requiring the Riemann Hypothesis is assumed or used. In Theorem 5.4, Assumption (E_{rem}) is explicitly labelled CONDITIONAL; it concerns the energy decomposition and is not a consequence of RH.

No GUE, Montgomery, or Hilbert–Pólya hypothesis. The Tehrani operator $\tilde{T} = \Phi \circ \Phi^*$ is a deterministic, explicitly computable finite-dimensional operator, constructed from the prime resonance vectors $\{a_p\}_{p \leq \kappa}$. It is not a random matrix, not a statistical ensemble, and not a Hilbert–Pólya operator. No GUE conjecture, no Montgomery pair correlation conjecture, and no random-matrix-theoretic assumption is used or implied anywhere in §§ 2–5. The eigenvector-localization pattern reported in § 3 is a numerical observation, not a postulate.

Status separation maintained. The zero ordinates γ_k appear as inputs to the construction of Φ , never as conclusions. The eigenvalues μ_j of $\tilde{T}$ are derived from Φ and are not identified with the γ_k .

Note on W_1 . The inverse operator $W_1 = C_T \cdot \tilde{T}^+$ does not assume RH or the Hilbert–Pólya postulate; however, its normalization constant C_T is calibrated against the zero ordinates

via OLS (§ 3.3). W_1 is therefore not fully input-free. $\tilde{T}$ itself is deterministic and explicitly computable.

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Call for Feedback

This paper is part of an open research initiative on prime–zero coupling and explicit-formula operators. All papers are freely available on Zenodo under CC BY 4.0, and the accompanying code is hosted on GitHub.

Zenodo:

https://zenodo.org/search?q=metadata.creators.person_or_org.name%3A%22Tehrani%2C%20Ulrich%22

GitHub:

<https://github.com/utehrani/analysislab-nt>

The author welcomes critical feedback, corrections, and collaboration. In particular: analytical approaches to Assumption (E_{rem}) (remainder control for $|\Delta_{\text{Cross}} + \Delta_{\text{Stream}}|$, Open Problem 6.2) and its relation to the Linear Independence conjecture and prime exponential sums, connections to existing results on exponential sums over primes and zeros of $\zeta(s)$, and any errors or gaps in the present arguments.

References

- [1] M.V. Berry and J.P. Keating, “ $H = xp$ and the Riemann zeros”, in: J.P. Keating, D.E. Khmelnitskii, I.V. Lerner (eds.), *Supersymmetry and Trace Formulae: Chaos and Disorder*, NATO ASI Series B: Physics, vol. 370, Plenum Press, New York, 1999, pp. 355–367.
- [2] E. Bombieri, “Problems of the Millennium: The Riemann Hypothesis”, Clay Mathematics Institute, 2000.
- [3] J.-B. Bost and A. Connes, “Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory”, *Selecta Mathematica, New Series* **1** (1995), no. 3, 411–457.
- [4] J.-F. Burnol, “Two complete and minimal systems associated with the zeros of the Riemann zeta function”, *J. Théor. Nombres Bordeaux* **16** (2004), no. 1, 65–94.
- [5] A. Connes, C. Consani, and H. Moscovici, “Zeta spectral triples”, arXiv:2511.22755 (2025).
- [6] A. Connes, “Trace formula in noncommutative geometry and the zeros of the Riemann zeta function”, *Selecta Math.* **5** (1999), no. 1, 29–106.
- [7] A. Connes and C. Consani, “Weil positivity and trace formula, the archimedean place”, *Sel. Math. (N.S.)* **27** (2021), Paper No. 77.
- [8] S. Endres and F. Steiner, “The Berry–Keating operator on $L^2(\mathbb{R}_+, dx)$ and on compact quantum graphs with general self-adjoint realizations”, *J. Phys. A: Math. Theor.* **43** (2010), 095204.
- [9] K. Ford, X. Meng, and A. Zaharescu, “Zeros of the Riemann zeta-function on the critical line and the distribution of the argument”, *Bull. London Math. Soc.* **49** (2017), 1–9.
- [10] A.E. Ingham, “On two conjectures in the theory of numbers”, *Amer. J. Math.* **64** (1942), 313–319.
- [11] H. Iwaniec and E. Kowalski, *Analytic Number Theory*, AMS Colloquium Publications 53, 2004.
- [12] M. Rubinstein and P. Sarnak, “Chebyshev’s bias”, *Experimental Math.* **3** (1994), 173–197.
- [13] D. Schumayer and D.A.W. Hutchinson, “Colloquium: Physics of the Riemann hypothesis”, *Rev. Mod. Phys.* **83** (2011), 307–330.
- [14] G. Sierra and J. Rodríguez-Laguna, “ $H = xp$ model revisited and the Riemann zeros”, *Phys. Rev. Lett.* **106** (2011), 200201.
- [15] G. Sierra and P.K. Townsend, “Landau levels and Riemann zeros”, *Physical Review Letters* **101** (2008), 110201.

- [16] P. Turán, “On some approximative Dirichlet-polynomials in the theory of the zeta-function of Riemann”, *Danske Videnskabernes Selskab, Mat.-fys. Medd.* **24** (1948), no. 17, 3–36 (36 pp.).
- [17] U. Tehrani, “A Curvature Decomposition of the Explicit Formula for the Riemann Zeta Function”, Zenodo, <https://doi.org/10.5281/zenodo.19025598>, 2026.
- [18] U. Tehrani, “From Local Curvature to the Weil Functional: An Explicit Construction”, Zenodo, <https://doi.org/10.5281/zenodo.19106992>, 2026.
- [19] U. Tehrani, “A Finite-Cutoff Hilbert-Space Model for Prime–Zero Energy Structure”, Zenodo, <https://doi.org/10.5281/zenodo.19307989>, 2026.